17 Mapping the Alpine Past: LiDAR Data Accessibility and Open-Source Approaches in the Belluno Dolomites

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Alpine landscapes are often presented as paradigmatic examples of the potential of LiDAR-derived Digital Elevation Models (DEMs) for archaeological research. Yet, despite the increasing availability of LiDAR data, coverage across Alpine regions remains incomplete or uneven, often because separate acquisition campaigns differ in resolution, accuracy, and data quality. Such heterogeneity poses significant challenges to the systematic investigation of archaeological landscapes, particularly at (sub)regional scales.

The Province of Belluno, in the Veneto Dolomites, offers an interesting case study. Although recently covered in its entirety by LiDAR surveys, the available datasets remain fragmented and technically heterogeneous, raising questions about their integration, processing, and reliability for archaeological interpretation. Moreover, access to these datasets is subject to a formal, justified request to the regional authorities, raising further questions about data accessibility and reuse.
This paper explores the opportunities and limitations of LiDAR-derived DEMs distributed by Regione Veneto for the identification and analysis of archaeological contexts, with particular attention to hilltop sites and upland areas. Drawing on workflows developed with open-source tools, particularly QGIS and the Relief Visualisation Toolbox (RVT), it discusses strategies for processing, integrating, and visualising heterogeneous datasets, while assessing the challenges of their archaeological interpretation.

More broadly, the paper reflects on the relationship between data accessibility and the adoption of open-source tools in archaeological research. It argues that the archaeological potential of regional LiDAR coverage depends on transparent, reproducible analytical workflows, but also on data availability and data-sharing policies that enable wider access to landscape-scale archaeological information.